

Prepare: Re-read FNRP Sections 2.3–2.4. Focus on how features of $\sigma_x(E)$ reveal the physics of the underlying reaction.

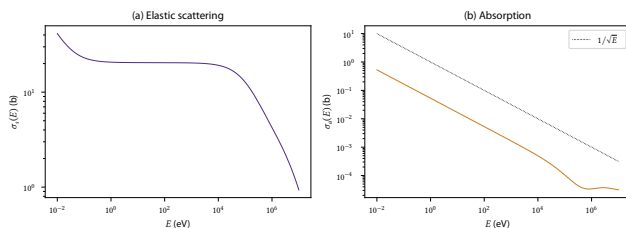
Primary objective

Access and inspect differential cross sections using OpenMC and map features of these cross sections to the underlying nuclear physics.

Vocabulary: resonance cross section; quantum state; energy level; energy spectrum; fission spectrum; Maxwell–Boltzmann distribution; $1/v$ cross section; threshold reaction; Breit–Wigner formula; E_r ; Γ ; Γ_n , Γ_γ ; wavelength λ ; Doppler broadening; random variable; probability density function (pdf); cumulative density function (cdf); expectation value

1 H-1: a simple place to start

Potential scattering and $1/v$ absorption



Low-energy capture

The capture cross section in the slide example follows

$$\sigma_\gamma(E) = \sqrt{\frac{E_0}{E}} \sigma_\gamma(E_0) \quad (\text{FNRP 2.36}).$$

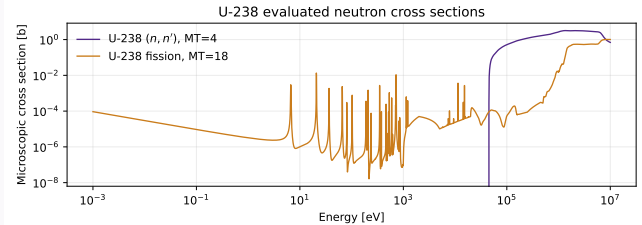
Thus $\sigma_\gamma \propto E^{-1/2} \propto 1/v$ for a nonrelativistic neutron.

$1/v$ check

If the neutron energy increases by a factor of 100, the $1/v$ capture cross section changes by a factor of _____. What feature of the H-1 elastic curve makes it qualitatively different?

2 Threshold reactions: a channel can turn on

²³⁸U threshold cross sections



What is a threshold telling us?

For a threshold channel x ,

$$\sigma_x(E) \approx 0, \quad E < E_{\text{th}},$$

$$\sigma_x(E) > 0, \quad E \gtrsim E_{\text{th}}.$$

On the plots, mark E_{th} and compare how ²³⁸U fission and inelastic scattering turn on.

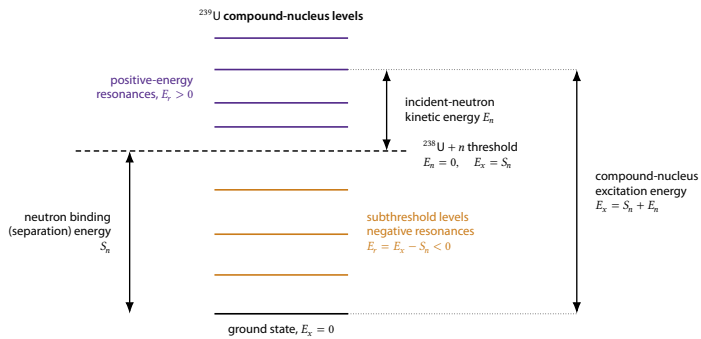
Connect shape to physics

A threshold is qualitatively different from a $1/v$ trend because the reaction channel is _____. What physical energy requirement must be met for inelastic scattering?

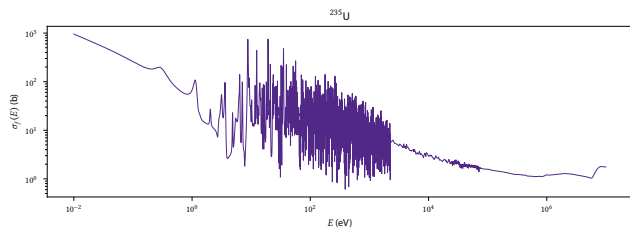
Reaction identifiers used in OpenMC

σ_t	1	Total cross section
σ_n, σ_e	2	Elastic scattering
$\sigma_{n'}, \sigma_{\text{in}}$	4, 51–91	Inelastic scattering
σ_{2n}	16	($n, 2n$)
σ_γ	102	Radiative capture (n, γ)
σ_f	18	Fission
σ_α	107	(n, α)
$\bar{\nu}$ (total)	452	Average total neutrons per fission
$\bar{\nu}_d$ (delayed)	455	Average delayed neutrons per fission
$\bar{\nu}_p$ (prompt)	456	Average prompt neutrons per fission

3 Resonances expose nuclear levels



^{235}U fission cross section



The fission cross section varies by several orders of magnitude over roughly ten decades of neutron energy. The jagged peaks between about 1 eV and 2 keV are **resonances**: a direct connection between the measured cross section and discrete states of the compound nucleus.

Read one resonance before writing a formula

Choose one resolved peak and identify:

location $E_r = \dots\dots\dots$

height $\sigma(E_r) \approx \dots\dots\dots$

width characteristic $\Gamma \approx \dots\dots\dots$

What does changing E_r do? What does changing Γ do?

Three useful shapes

Across the examples in this lesson, recognize at sight:

- a smooth low-energy $1/v$ trend,
- a reaction threshold,
- a resonance peak associated with a nuclear level.

4 Single-level Breit-Wigner (SLBW)

Capture near one resonance

For a resonance centered at E_r ,

$$\sigma_\gamma(E) = \sigma_0 \frac{\Gamma_\gamma}{\Gamma} \left(\frac{E_r}{E} \right)^{1/2} \frac{1}{1 + 4(E - E_r)^2 / \Gamma^2} \quad (\text{FNRP 2.41})$$

Elastic scattering near one resonance

$$\sigma_n(E) = \sigma_0 \frac{\Gamma_n}{\Gamma} \frac{1}{1 + 4(E - E_r)^2 / \Gamma^2} + \sigma_0 \frac{2R}{\lambda_0} \frac{2(E - E_r)/\Gamma}{1 + 4(E - E_r)^2 / \Gamma^2} + 4\pi R^2. \quad (\text{FNRP 2.42})$$

What do E_r and Γ control?

For the Lorentzian factor

$$L(E) = \frac{1}{1 + 4(E - E_r)^2 / \Gamma^2},$$

complete the two checks:

$$L(E_r) = \dots\dots\dots, \quad L(E_r \pm \Gamma/2) = \dots\dots\dots.$$

Hence E_r locates the peak and Γ sets its characteristic width. How do Γ_γ/Γ and Γ_n/Γ partition the resonance between channels?

Many resonances

To reconstruct a cross section over a region containing several resonances, evaluate the single-level expression for each resonance—with its own E_r , σ_0 , Γ_γ , and Γ_n —and sum the contributions.

Minimal OpenMC lookup

For a nuclide loaded as data, a reaction is selected by its MT number, and its temperature-dependent cross section is a callable function of energy:

```
# Once OpenMC is loaded, you can use the following.
xs_path = os.environ.get("OPENMC_CROSS_SECTIONS")
xs_path = xs_path.replace("cross_sections.xml", "neutron/")
u235_fission = U235[18].xs["294K"]
u235_capture = U235[102].xs["294K"]
```

For example, MT 18 selects fission and MT 102 selects radiative capture.